GastroOps: REST API Reference Manual

FastAPI Microservice Endpoints • Multi-Agent Protocols • Payload Schemas

1. System Architecture Overview

The GastroOps API is built on high-performance FastAPI (ASGI) and coordinates multi-stage hybrid retrieval, 8 specialized autonomous agents, classical supervised machine learning, and on-demand multilingual localization. All microservice endpoints operate over standard HTTP/JSON and include automatic Swagger UI documentation at /docs.

Service	Base URL / Port	Protocol	Purpose
FastAPI Backend	http://127.0.0.1:8000	REST / JSON	Core multi-agent analysis, search, translation
React Web UI	http://127.0.0.1:3000	HTTP / HTML5	Manager dashboard & interactive telemetry
Qdrant Vector DB	http://127.0.0.1:6333	gRPC / REST	BGE-M3 1024-dim dense vector search engine
PostgreSQL 14	127.0.0.1:5432	TCP / SQL	Yelp review corpus & business metadata store

2. Core Microservice Endpoints Specification

[POST] /api/search — Multi-Stage Hybrid Review Search & Synthesis

Executes 5-stage hybrid search (BGE-M3 Dense + BM25 Sparse + RRF + BGE Reranker v2 M3 + Evidence Gate) with polarity star boundaries and 1-paragraph executive synthesis.

Request Payload:

```
{
  "query": "slow service and cold food",
  "alpha": 0.5,
  "top_k": 3,
  "business_id": "all",
  "min_stars": 1
}
```

Response Structure:

```
{
  "query": "slow service and cold food",
  "total_retrieved": 3,
  "results": [ ... ],
  "executive_summary": "An analysis of customer reviews reveals...",
  "key_reasons": ["Food temperature control lapses", "Kitchen ticket pacing bottlenecks"],
  "diagnostics": { "evidence_gate_status": "sufficient_evidence" }
}
```

[POST] /api/multi-agent — Full 8-Agent Collaborative Operations Pipeline

Executes full multi-agent workflow: Input Guardrails, Query Understanding, Hybrid Retrieval, Taxonomy Classification, Prioritization (P0-P4), Forensic Root-Cause ABSA, Operations Strategist SOPs, and LLM-as-Judge Audit.

Request Payload:

```
{
  "query": "raw chicken and violent illness",
  "top_k": 3
}
```

Response Structure:

```
{
  "success": true,
  "classification": { "primary_category": "Cleanliness & Hygiene", "urgency": "Critical" },
  "prioritization": { "priority_tier": "P0 - Critical", "priority_score": 92.0, "action_color": "red" },
  "recommendations": {
    "immediate_actions": [ "Immediate protein batch quarantine and HACCP safety hold" ],
    "preventative_sops": [ "SOP-QA-100: Mandatory digital temperature probe verification" ],
    "success_kpis": [ { "metric": "Food Safety Incidents", "target": "0 complaints" } ]
  },
  "judge_audit": { "verdict": "APPROVED", "overall_score_pct": 88 }
}
```

[POST] /api/evaluate — DeepEval Recommendation Quality Audit

Audits recommendation action plans against 4 DeepEval dimensions: Faithfulness, Answer Relevancy, Aspect Coverage, and Operational Feasibility.

Request Payload:

```
{
  "query": "Kitchen ticket delays and undercooked entrees",
  "alpha": 0.5
}
```

Response Structure:

```
{
  "deepeval_suite": {
    "overall_score_pct": 86.1,
    "status": "PASSED (Production Grade)",
    "test_results": [
      { "metric": "Faithfulness", "score": 0.88, "passed": true },
      { "metric": "Answer Relevancy", "score": 0.89, "passed": true },
      { "metric": "Aspect Coverage", "score": 0.67, "passed": true },
      { "metric": "Operational Feasibility", "score": 0.98, "passed": true }
    ]
  }
}
```

[POST] /api/translate — On-Demand Multilingual Localization

Translates the English executive summary and reasons into Kannada (kn), Hindi (hi), Telugu (te), or Tamil (ta). Strictly preserves proper nouns, numbers, and SOP codes verbatim.

Request Payload:

```
{
  "summary": "Customer complaints at Bella Roma Trattoria center on cold food.",
  "reasons": [ "Improper warming station temperatures" ],
  "target_language": "kn"
}
```

Response Structure:

```
{
  "success": true,
  "target_language": "kn",
  "summary": "Bella Roma Trattoria ■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■ ■■■■■ ■■■■■■■■■■■■■■■■■■...
",
  "reasons": [ "■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■ ■■■■■■■" ],
  "cached": false
}
```

[POST] /api/csat/predict — Supervised ML CSAT & Star Rating Prediction

Uses scikit-learn TF-IDF + Ridge Regression to predict customer satisfaction percentages (0-100%) and star ratings (1.0-5.0) directly from review text.

Request Payload:

```
{
  "text": "The soup was freezing cold and service took an hour. Terrible experience."
}
```

Response Structure:

```
{
  "predicted_stars": 1.2,
  "predicted_csat_pct": 5.0,
  "nps_category": "Detractor",
  "model_architecture": "TF-IDF N-Gram Vectorizer + Ridge L2 Regression"
}
```

[GET] /api/trends — Time-Series Complaint Velocity & Slope Forecasting

Aggregates historical monthly complaint series and calculates linear regression trajectory slopes via NumPy polyfit, forecasting next-month complaint surges.

Request Payload:

```
GET /api/trends?category=Food%20Quality
```

Response Structure:

```
{
  "trend_slope": 0.42,
  "overall_trajectory": "SURGING (Urgent Alert)",
  "projected_next_month_complaints": 24,
  "historical_monthly_data": [ ... ]
}
```